Bryan Chang

bryan.chang@berkeley.edu | 217-200-2182 | linkedin.com/in/bryanchang9 | github.com/Bryan1203

Objective

Robotics engineer specializing in autonomous systems with expertise in LiDAR-based mapping, perception, and localization. Seeking to develop robust navigation solutions for autonomous vehicles through advanced sensor fusion and SLAM algorithms.

Education

University of California, Berkeley | *M.Eng. Electrical Engineering & Computer Science* | May 2026
GPA: 3.78/4.0 | Specialization: Robotics and Embedded Software | UC Berkeley Fung Fellowship Recipient

University of Illinois at Urbana-Champaign | *B.S. Computer Engineering with Honors* | Dec 2024
GPA: 3.79/4.0 | Dean's List: Spring 2022, Fall 2022

Technical Skills

Languages: Python, C++, C, CUDA | **Tools:** ROS/ROS2, PyTorch, OpenCV, PCL, MoveIt, Docker, Git
Specializations: SLAM, LiDAR Processing, Motion Planning, Deep Learning, Sensor Fusion, Embedded Systems

Work Experience

Autonomous Driving Software Engineer Intern | *Industrial Technology Research Institute* | Summer 2024, 2025

- Reduced LiDAR lane annotation time by 97% (2 weeks → 1 hour) through deep learning automation, enabling rapid dataset iteration and model training cycles for autonomous driving perception systems
- Improved 3D lane detection accuracy from 68% to 97% through intensity-based augmentation and semantic segmentation; enhanced dashed lane recall from 76% to 91% via multi-frame temporal fusion for highway navigation
- Developed custom annotation pipeline with visualization tools for 1,000+ manually labeled LiDAR scans with quality metrics
- Integrated Hesai AT128 LiDAR (128 channels, 200m range) into production stack; validated across urban, highway, and rural environments with HD map alignment

Internet of Things Undergraduate Teaching Assistant | *University of Illinois* | Fall 2024

- Developed capstone showcase platform (Link) featuring 22 student projects; mentored 60+ students in IoT system design and embedded programming

Key Technical Projects

High-Precision LiDAR-Inertial Localization | *AI Racing Tech, UC Berkeley* | Jan 2025 – Present

- Developing ROS2 localization system based on Direct LiDAR-Inertial Odometry (DLIO) framework with GLIM mapping and GNSS constraints for centimeter-level prior map generation across 2km race circuit
- Implemented real-time pose estimation pipeline using 100Hz IMU pre-integration with Kalman filtering and GICP scan matching against prior map for drift correction
- Achieved 50cm mean localization error vs RTK-GNSS ground truth (10cm baseline accuracy) through sensor fusion, enabling precise trajectory following at racing speeds up to 100 mph on Indy Autonomous Challenge vehicle

LiDAR-Based Lane Navigation | *Project Lead* | Sep – Dec 2024 | GitHub

- Engineered end-to-end ROS autonomous system processing 10Hz LiDAR point clouds (100k+ points/frame) through Point Transformer V3 neural network for semantic lane detection, with KISS-ICP SLAM for temporal fusion addressing sensor blind spots
- Achieved 350ms average latency through optimized PyTorch inference with CUDA acceleration and asynchronous processing; implemented buffered mapping for robustness against sensor dropouts
- Deployed Pure Pursuit controller with adaptive PID on Polaris GEM e2/e4 vehicles, navigating 500m test course with 20cm lane-keeping accuracy at speeds up to 15 mph

Robotic Pick-and-Place Automation | *Ember Robotics, UC Berkeley* | Jan – Mar 2025 | Website

- Programmed Techman TM12 6-DOF robot for automated glass slide handling using inverse kinematics and MoveIt in ROS2
- Achieved 0.5mm positioning accuracy with 95% success rate over 200+ cycles, reducing sample preparation time by 60%

Firefighter Health Monitoring Network | *Hardware Lead* | Fall 2024 | GitHub

- Designed ESP32-based wearable system with 300m+ mesh network range in urban environments for real-time vital signs monitoring
- Achieved 2+ hour battery life at high temperatures; built real-time GUI for incident commanders to track multiple firefighters
- *Honorable Mention & Hall of Fame - one of five teams awarded out of 45 total teams*

Bicycle-Mounted Pothole Detection System | *Project Lead* | Spring 2024 | Report

- Engineered real-time pothole detection using custom YOLO model on Raspberry Pi with Edge TPU acceleration
- Accelerated inference speed by 40% through quantization and 837% via TPU, achieving real-time performance on edge device
- Developed Bluetooth hazard warning network with voice alerts to rider and custom taillight signals for approaching cyclists